CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 13: Factorisation

Topic-wise short notes with solved examples

Factorising algebraic expressions by taking out common factors, by grouping, by using identities and by splitting the middle term of a trinomial.

1 Factors

- To **factorise** an expression is to write it as a **product** of simpler expressions. Each one is a **factor**.
- Factorising is the reverse of multiplying: $3x(x+2) = 3x^2 + 6x$, so $3x^2 + 6x = 3x(x+2)$.
- Check any factorisation by multiplying the factors back.
- **Always take out the common factor first**, then look at what is left.

2 Common factors

- Find the HCF of the terms: the HCF of the numerical coefficients and the lowest power of each common variable.
- Write the HCF outside a bracket and divide each term by it.

Example 2.1 Factorise $6x^2y - 9xy^2 + 3xy$.

SOLUTION

HCF $= 3xy$

$3xy(2x - 3y + 1)$

Example 2.2 Factorise $5(a+b) - 2x(a+b)$.

SOLUTION

The bracket $(a+b)$ is common.

$(a+b)(5-2x)$

Example 2.3 Factorise $x(a-3) + y(3-a)$.

SOLUTION

$3 - a = -(a-3)$, giving $x(a-3) - y(a-3)$.

$(a-3)(x-y)$

3 Grouping

- Used for four (or more) terms with no factor common to all.

- Arrange the terms in groups so that each group has a common factor and the same bracket is left in each.

Example 3.1 Factorise $ab + bc + ax + cx$.

SOLUTION

$b(a + c) + x(a + c)$

$(a + c)(b + x)$

Example 3.2 Factorise $x^2 - 3x - xy + 3y$.

SOLUTION

$x(x - 3) - y(x - 3)$

$(x - 3)(x - y)$

Example 3.3 Factorise $6ab - 4b + 6 - 9a$.

SOLUTION

$2b(3a - 2) - 3(3a - 2)$

$(3a - 2)(2b - 3)$

4 Difference of two squares

$$a^2 - b^2 = (a + b)(a - b)$$

- Each term must be a perfect square and the sign between them must be minus.
- After using the identity, check whether a factor can be factorised again.

Example 4.1 Factorise $49x^2 - 81y^2$.

SOLUTION

$(7x + 9y)(7x - 9y)$

Example 4.2 Factorise $3a^3 - 48a$.

SOLUTION

$3a(a^2 - 16)$

$3a(a + 4)(a - 4)$

Example 4.3 Factorise $x^4 - 1$.

SOLUTION

$(x^2 + 1)(x^2 - 1)$

$(x^2 + 1)(x + 1)(x - 1)$

Example 4.4 Factorise $(a + b)^2 - c^2$.

SOLUTION

$(a + b + c)(a + b - c)$

Example 4.5 Evaluate $(73)^2 - (27)^2$.

SOLUTION

$(73 + 27)(73 - 27) = 100 \times 46$

4600

5 Perfect square trinomials

- $a^2 + 2ab + b^2 = (a + b)^2$ and $a^2 - 2ab + b^2 = (a - b)^2$.
- **Test**: the first and last terms are perfect squares and the middle term is twice the product of their square roots.

Example 5.1 Factorise $9x^2 + 24xy + 16y^2$.

SOLUTION

$(3x)^2 + 2(3x)(4y) + (4y)^2$

$(3x + 4y)^2$

Example 5.2 Factorise $4a^2 - 20a + 25$.

SOLUTION

$(2a - 5)^2$

Example 5.3 Factorise $x^2 + 6x + 9 - y^2$.

SOLUTION

$(x + 3)^2 - y^2$

$(x + 3 + y)(x + 3 - y)$

6 Trinomials of the form x² + bx + c

- Find two numbers whose **product is** c and whose **sum is** b. If they are p and q, then $x^2 + bx + c = (x + p)(x + q)$.

Sign of c	Sign of b	Signs of p and q
$+$	$+$	Both positive
$+$	$-$	Both negative
$-$	$+$	Opposite signs; the larger is positive
$-$	$-$	Opposite signs; the larger is negative

Example 6.1 Factorise $x^2 + 9x + 20$.

SOLUTION

Product 20, sum 9: 4 and 5.

$(x + 4)(x + 5)$

Example 6.2 Factorise $x^2 - 7x + 12$.

SOLUTION

Product 12, sum -7: -3 and -4.

$(x - 3)(x - 4)$

Example 6.3 Factorise $x^2 + 2x - 15$.

SOLUTION

Product -15, sum 2: 5 and -3.

$(x + 5)(x - 3)$

Example 6.4 Factorise $y^2 - 4y - 21$.

SOLUTION

Product -21, sum -4: -7 and 3.

$(y - 7)(y + 3)$

7 Trinomials of the form ax² + bx + c

- Find two numbers whose **product is** ac and whose **sum is** b. Split the middle term with them and factorise by grouping.

Example 7.1 Factorise $2x^2 + 7x + 3$.

SOLUTION

Product 6, sum 7: 6 and 1. $2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3)$.

$(x + 3)(2x + 1)$

Example 7.2 Factorise $3x^2 - 10x + 8$.

SOLUTION

Product 24, sum -10: -6 and -4. $3x^2 - 6x - 4x + 8 = 3x(x - 2) - 4(x - 2)$.

$(x - 2)(3x - 4)$

Example 7.3 Factorise $6a^2 - a - 2$.

SOLUTION

Product -12, sum -1: -4 and 3. $6a^2 - 4a + 3a - 2 = 2a(3a - 2) + 1(3a - 2)$.

$(3a - 2)(2a + 1)$

8 Using factors to divide

- To divide one expression by another, factorise both and cancel the common factors.

Example 8.1 Divide $x^2 - 5x + 6$ by $x - 2$.

SOLUTION

$x^2 - 5x + 6 = (x - 2)(x - 3)$

$x - 3$

Example 8.2 Simplify $\dfrac{4x^2 - 9}{2x^2 + 3x}$.

SOLUTION

$$\frac{(2x + 3)(2x - 3)}{x(2x + 3)}$$

$$\frac{2x - 3}{x}$$

9 Common mistakes

- Leaving a common factor inside the bracket.
- Sign errors when the second group starts with a minus.
- Trying to factorise a sum of squares such as $x^2 + 9$.
- For $ax^2 + bx + c$, looking for numbers with product c instead of ac.
- Stopping at $(x^2 + 1)(x^2 - 1)$ without factorising $x^2 - 1$.
- Not checking by multiplying the factors back.

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